Daniël Trujillo

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Education

Massachusetts Institute of Technology (MIT) — Cambridge, MA, USA
PhD in EECS, CSAIL. Advisor: Mengjia Yan — *Sept. 2023 – Early 2028 (anticipated)*

ETH Zürich — Zürich, Switzerland
*MSc in Computer Science. GPA 5.49 / 6.00 (**ETH Medal**)* — *Sept. 2020 – June 2023*

Vrije Universiteit (VU) Amsterdam — Amsterdam, Netherlands
*BSc in Computer Science. GPA: 9.3 / 10.0 (**Cum Laude**)* — *Sept. 2017 – Aug. 2020*

Publications *(* = joint first authorship)*

TONTOU: On the Exploitability of Time-of-Neutralization to Time-of-Use Windows — Aug. 2026
Daniël Trujillo, Mengjia Yan
USENIX Security 2026

Defeating Transient Execution Attacks by Limiting Secret Reachability through Register Hiding and ShadowCFI — May 2026
Daniël Trujillo, Jagadish Kotra, David Kaplan, Mengjia Yan
IEEE S&P 2026

Phantom: Exploiting Decoder-detectable Mispredictions *(Best Paper Award)* — Oct. 2023
Johannes Wikner*, Daniël Trujillo*, Kaveh Razavi
MICRO 2023

Inception: Exposing New Attack Surfaces with Training in Transient Execution — Aug. 2023
Daniël Trujillo*, Johannes Wikner*, Kaveh Razavi
USENIX Security 2023

TLB;DR: Enhancing TLB-based Attacks with TLB Desynchronized Reverse Engineering — Aug. 2022
Andrei Tatar*, Daniël Trujillo*, Cristiano Giuffrida, Herbert Bos
USENIX Security 2022

CVEs

CVE-2023-20569 (Inception / "SRSO") — Aug. 2023
AMD

CVE-2022-23825 (Phantom / "Branch Type Confusion") — Oct. 2023
AMD

Experience

NVIDIA — May 2026 – Aug. 2026
PhD Research Intern, Security and Privacy — *(upcoming)*

Microsoft Research — June 2025 – Aug. 2025
Research Intern, Software HARdware Co-design (SHARC) — *Redmond, WA, USA*
- Researched exposing memory access internals to software
- Developed new mechanisms enabling safe memory de-duplication and reducing memory-access overheads

AMD — June 2024 – Jan. 2025
Research Intern, Research and Advanced Development (RAD) — *Austin, TX, USA*
- Researched more performant mitigations against transient execution attacks
- Contributed to software-only defenses for the Linux kernel, reducing overhead

Intel Corporation — July 2022 – Nov. 2022
Offensive Security Research Intern, Intel Product Assurance and Security (IPAS) — *Zürich, Switzerland*
- Contributed to taint analysis prototype, specialized in finding stale data bugs semi-automatically
- Demonstrated existence of a known stale bug in Intel IP

IBM Research — June 2021 – Sep. 2021
Intern, Cloud Data Platforms — *Zürich, Switzerland*
- Developed PoC of software abstraction of smartNIC storage-offloading functionalities

Awards & Honors

Intel Bug Bounty Program Bonus — March 2026
Intel Corporation

NDSEG Fellowship — Sep. 2024 – Aug. 2027
U.S. Department of Defense

Best Paper Award — Oct. 2023
MICRO 2023

ETH Medal — Sep. 2023
ETH Zürich

Best Hack & Use of Google Cloud — Oct. 2019
HackHarvard

GLOBE Scholarship — Sep. 2019
VU Amsterdam

Patents

Enabling secure memory sharing on cloud servers — Jan. 2026
Daniël Trujillo, Liangcheng Yu, Stefan Saroiu, Alec Wolman — *(Pending)*
Microsoft

Custom-sized pages for reducing TLB misses — Jan. 2026
Daniël Trujillo, Liangcheng Yu, Stefan Saroiu, Alec Wolman — *(Pending)*
Microsoft

Fine-grained cache hit/miss profiling — Jan. 2026
Daniël Trujillo, Liangcheng Yu, Stefan Saroiu, Alec Wolman — *(Pending)*
Microsoft

System and method for preventing transient execution attacks — Jun. 2025
Jagadish Kotra, Daniël Trujillo, David Kaplan — *(Pending)*
AMD

Detecting potential security issues in a hardware design using RTL information flow tracking — Mar. 2023
Ben Gras, Daniël Trujillo — *(Pending)*
Intel Corporation

Selected Talks

TONTOU: Weaponizing Interrupts to Break Spectre Defenses on AMD and Intel — Aug. 2026
Daniël Trujillo — *Las Vegas, NV, USA*
Black Hat USA

From Phantom to Inception — May 2024
Daniël Trujillo, Johannes Wikner — *Berlin, Germany*
OffensiveCon

Inception — Dec. 2023
Daniël Trujillo — *São Paulo, Brazil*
H2HC

TLB;DR — Aug. 2022
Daniël Trujillo — *Boston, MA, USA*
USENIX Security

Teaching Experience

ETH Zürich — Jan. – May 2022
Teaching Assistant — Computer Engineering — *Zürich, Switzerland*

VU Amsterdam — June 2018 – May 2020
Teaching Assistant — *Amsterdam, Netherlands*

- Programming for Economists in Python (Jun 2018 & Jun 2019)
- Computer Programming in C++ (Sept – Oct 2018)
- Physical Computing (Nov – Dec 2018)
- Web Technology (Jan 2019 & Jan 2020)
- Pervasive Computing (Feb – Mar 2019)
- Natural Language Processing Technology (Apr – May 2020)